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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 215 — Past Questions & Materials
by Renance

Course: MTH 215
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200

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Paper 1 of 5 — MTH 215 Vector Analysis First Semester 2020 2021

Level 200 • First Semester • PastQuestion

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2ST SEMESTER EXAMINATION 2020/2021

Tae meconne Code/Title: MTH 215 — Vector Analysis

ions: Attempt any 4 questions. Time Allowed: 2 Hours.

la) Define vector equation of a plane

b) Determine th . aca

Pah ne the equation of the plane $X(s,t) = P + sA + tAifP = (1,2,-3)$

$= (3,2,1)$ and $B = (1,0,4)$. Determine which of these points is on the plane

$(6,4,6)$ and $(1,2,0)$ (6.5 marks)

C } |

) A Particle moves along a curve $xX = 2t^2, y = t^* - 4t$, and $z = 3t - 5$, Find the component of . velocity and acceleration att = 1, in the direction $i - 3j + 2k$. (6 marks)

(6 marks)

f vector algebra.

d the angle between the vectors a and

2a) Define Vector Algebra and State five laws 0

b) Given that $a = -it + 2j = 2k$, $b = 3i + 4j - 12k$, Fin

b. (5 marks)

C) $fA = -it + 2k, B = 2it + j - *$ and $C = it + 2j + 2k$. Compute each of the following: (1)

$AX(BXC)$ (ii) $(At B)X(A-C)$ (iii) $(AXB)X(AXC)$ (6.5 marks)

3a) Find the equation of: (i) Tangent line (ii) Normal line to the surface

$x? + y^* - 47$ at $(2, -4, 5)$ is marks)

b) Find the shortest distance from the point $(1, 1, -2)$ to the surface $2 = x^* + y$ (5 marks)

c) Find the Curl and Divergence of the vector $V = 3x^2zi + xyzj + TYk$ (5 marks)

4a) Find the directional directive of the scalar field $u = XY^4 + yz + 2x$ al the point

$(1,1,1)$ in the direction and normal to the plane $x + r + 2y + 2z = 7$ (7.5 marks)

b) Prove that $VX VG = Oif O's$ assumed to be continuous and having continuous

First and second partial derivatives | (5 marks)

c) $fy = xeyl - 2y^*zj + xy?z^*k$. Find $= -$ at the point $(2, 1, -2)$. (5 marks)

. 5a) Let F be a two-dimensional vector field given by $F(x,y) = jyi + (x? + y)j$ for all $(x,y) \in \mathbb{R}^2$:

Calculate the line integral of F from $(0,0)$ to $C(1, 1)$ along each of the following paths: (i) The line

'ons $x = ty = b$ $0 < t < 1$ (ii) the path with parametric equations

(10 marks)

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{ . $F dr = j, |, / y dx + (x^3 + y) ay]$

b) Evaluate $\oint_C \mathbf{A} \cdot d\mathbf{s}$ where $\mathbf{A} = x\mathbf{i} - 3y^2z\mathbf{k}$ and S is the surface of cylinder $x^2 + y^2 = 16$ included in the First octant between $z = 0$ and $z = 9$ (7.5 marks)

for $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$,

Find $\nabla \phi$ at the point $(2, 1, -2)$ (7 marks)

$\mathbf{a} = x\mathbf{i} + y\mathbf{j}$ |

$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector of a point on the space curve,

1

1) $\mathbf{r} = (x^2, 2z)$ (5 marks)

find V if $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$)

(5.5 marks)

1 .

22, Find V where $Q(r)$ is a function of r alone.

Si

$p(x, y, z) =$

Paper 2 of 5 — MTH 215 Vector Analysis First Semester 2020 2021

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2ST SEMESTER EXAMINATION 2020/2021

Tae aeoe Code/Title: MTH 215 — Vector Analysis

ions: Attempt any 4 questions. Time Allowed: 2 Hours.

1a) Define vector equation of a plane

b) Determine the equation of the plane $X(s,t) = P + sA + tB$ if $P = (1,2,-3)$

$A = (3,2,1)$ and $B = (1,0,4)$. Determine which of these points is on the plane $(6,4,6)$ and $(1,2,0)$ (6.5 marks)

2) A Particle moves along a curve $x = 2t^7$, $y = +2 - At$, and $z = 3t - 5$, Find the component of - velocity and acceleration at $t = 1$, in the direction $t - 3j + 2k$. (6 marks)

2a) Define Vector Algebra and State five between the vectors a and b

b) Given that $a = -i + 2j + 2k$, $b = 3i + 4j - k$. Compute each of the following: (i) $A \times (B \times C)$ (ii) $(A + B) \times (A - C)$ (iii) $(A \times B) \times (A \times C)$ (6.5 marks)

3a) Find the equation of: (i) Tangent line (ii) Normal line to the surface $x^2 + y^2 = 4z^2$ at $(-4, 5)$ (7.5 marks)

b) Find the shortest distance from the point $(1, 1, -2)$ to the surface $Z = x^2 + y^2$ (5 marks)

c) Find the Curl and Divergence of the vector $V = 3x^2z^2i + xyzj + y^2k$. (5 marks)

4a) Find the directional derivative of the scalar field $u = xy + yz$ at the point $(1,1,1)$ in the direction and normal to the plane $x + 2y + 2z = 7$ (7.5 marks)

b) Prove that $\nabla \cdot (\nabla \times V) = 0$ if V is assumed to be continuous and having continuous first and second partial derivatives. (5 marks)

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c) $f_y = x^2y - 2y^2$ at the point $(2, 1, -2)$. (5 marks)

5a) Let F be a two-dimensional vector field given by $F(x,y) = (y^2 + x^2)j$ for all $(x,y) \in \mathbb{R}^2$.

Calculate the line integral of F from $(0,0)$ to $(1, 1)$ along each of the following paths: (i) The line with parametric equations: $x = t, y = t, 0 \leq t \leq 1$ (ii) the path with parametric equations

$[x^2, y^2, z^2]$ (10 marks)

b) Evaluate $\iint_S A \cdot dS$ where $A = xj - 3y^2zk$ and S is the surface of cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 9$. (7.5 marks)

6a) if $r = x^2i + 2yzj + xy^2k$,

find $\oint_C r \cdot dr$ (7 marks)

Find $\frac{\partial^2 z}{\partial x^2}$ at the point $(2, 1, 2)$ if

$r = x^2i + y^2j + z^2k$ is the position vector of a point on the space curve,

$r = t^2i + t^3j + t^4k$

2. Find

and $v(2) = \frac{d}{dt} r(t) \cdot z^*$: (5 marks)

$r = t^2i + t^3j + t^4k$

(5.5 marks)

2. Find VQ , where $Q(r)$ is a function of T alone.

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ep $a(x)$ is

Paper 3 of 5 — MTH 215 Vector Analysis First Semester 2020 2021

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2ST SEMESTER EXAMINATION 2020/2021

Tae aeoe Code/Title: MTH 215 — Vector Analysis

ions: Attempt any 4 questions. Time Allowed: 2 Hours.

1a) Define vector equation of a plane

b) Determine th ae:

: ine the equation of the plane $X(s,t) = P + sA + tA$ if $P = (1,2,-3)$

$= (3,2,1)$ and $B = (1,0,4)$. Determine which of these points is on the plane

$(6,4,6)$ and $(1,2,0)$ (6.5 marks)

) A Particle moves along a curve $x = 2t^7$, $y = +2 - At$, and $z = 3t - 5$, Find the component of - velocity and acceleration at $t = 1$, in the direction $t - 3j + 2k$. (6 marks)

(6 marks)

2a) Define Vector Algebra and State five

between the vectors @ and

b) Given that $a = -i + 2j = 2k, b = 3i + 4j - k$

b. (5 marks)

C) if $A = -i + 2k$, $B = 2i + j - k$ and $C = i + 2j + 2k$. Compute each of the following: (i)

$A \times (B \times C)$ (ii) $(A + B) \times (A - C)$ (iii) $(A \times B) \times (A \times C)$ (6.5 marks)

3a) Find the equation of: (i) Tangent line (ii) Normal line to the surface

$x^2 + y^2 = 4z^2$ @ $(-4, 5)$ (7.5 marks)

b) Find the shortest distance from the point $(1, 1, -2)$ to the surface $Z = x^2 + y^2$ (5 marks)

c) Find the Curl and Divergence of the vector $V = 3x^2z^2i + xyzj + y^2k$. (5 marks)

4a) Find the directional derivative of the scalar field $u = xy + yz$ at the point $(1, 1, 1)$ in the direction and normal to the plane $x + 2y + 2z = 7$ (7.5 marks)

b) Prove that $\nabla \cdot (\nabla \times V) = 0$ if V is assumed to be continuous and having continuous first and second partial derivatives. (5 marks)

; a^2

c) $f_y = x^2y - 2y^2$ $f_{xy} = f_{yx}$. Find f at the point $(2, 1, -2)$. (5 marks)

5a) Let F be a two-dimensional vector field given by $F(x,y) = (y^2 + x^2)j$ for all $(x,y) \in \mathbb{R}^2$.

Calculate the line integral of F from $(0,0)$ to $(1, 1)$ along each of the following paths: (i) The line with parametric equations: $x = t, y = t, 0 \leq t \leq 1$ (ii) the path with parametric equations

$[F \cdot dr = f_1 dx + f_2 dy + f_3 dz]$ (10 marks)

b) Evaluate $\iint_S A \cdot dS$ where $A = xj - 3y^2zk$ and S is the surface of cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 9$. (7.5 marks)

6a) if $r = x^2i + 2yzj + xy^2k$,

find $\oint_C r \cdot dr$ (7 marks)

Find $\frac{d}{dt} (x^2y^2)$ at the point $(2, 1, 2)$ if

$r(t) = x(t)i + y(t)j + z(t)k$ is the position vector of a point on the space curve,

$r(1) = 2i + j + 2k$

$r'(1) = 2i + j$

and $v(2) = \frac{1}{2}r'(2)$: (5 marks)

$r(0) = i + j + k$

(5.5 marks)

Find $\frac{d}{dt} (x^2y^2)$, where $r(t)$ is a function of t alone.

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ep $a(x, y, z)$ is

Paper 4 of 5 — MTH 215 2016 2017 EXAM Apst.Sam.

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2016/2017
Course Code: MTH 215 **Title: Statistical Methods**
Level: 200

Instruction: Answer any four questions. **Duration: 2½ hrs**

- (a) Explain the reasons for designing an experiment

(b) A survey is designed to determine the mean amount of time corporation executives watch television. A pilot survey indicated that the mean time per week is 12 hours with a standard deviation of half an hour. It is desired to estimate the mean viewing time within a margin of a quarter of an hour. How many executives should be surveyed?

(c) Two different test methods are used for determining the viscosity of synthetic rubber. One method is used in the manufacturer's laboratory and the other is used in the customer's lab. Ten test samples were split with one part of the sample sent to each lab. The following data were obtained:

Manufacturer	91.70	91.87	93.00	93.94	92.61	92.89	92.63	92.35	92.21	93.44
Customer	92.26	96.59	95.05	93.66	94.61	95.60	93.44	93.84	93.76	96.17

Test at $\alpha = 0.05$ whether the two test methods are equivalent. Based on your analysis what type of customer-supplier interactions can you anticipate?
- (a) Explain the following statistical concepts:

(i) Replication (ii) Randomization (iii) Factor (iv) Treatments

(b) Two petroleum-product refineries produce batches of fuel using the same refining process. A random sample of batches at each refinery resulted in the following values of cetane number measured on the fuel samples. Test the significant difference of the mean cetane number measured from the two refineries at $\alpha = 0.01$.

Refinery 1: 50.36, 47.54, 47.63, 47.67, 47.78, 47.29, 47.55, 48.21, 46.95, 47.95, 47.35, 48.37, 49.27, 47.23, 48.69, 47.57, 49.10, 49.88, 48.70, 47.91.

Refinery 2: 45.83, 45.90, 45.30, 45.45, 46.13, 46.69, 45.63, 46.15, 46.32, 45.22, 46.20, 45.50, 45.75, 45.57.

(c) Using the data in 2 (b), test the significant difference of the two sample means using Mann-Whitney test.
- (a) In an experiment to determine the effect of temperature on the lifetime (in hours) of an encapsulating resin for gold-aluminium bonds in integrated circuits. Thirty-seven units were assigned at random to one of five different temperature stresses. The Table below gives the log10 lifetimes in hours for the test units. Test the significant of the temperature stresses on resin bond lifetime at $\alpha = 0.05$.

Temperature (Degree Centigrade)									
175		194		213		231		250	
2.04	1.85	1.66	1.66	1.53	1.35	1.15	1.21	1.26	1.02
1.91	1.96	1.71	1.61	1.54	1.27	1.22	1.28	0.83	1.09
2.00	1.88	1.42	1.55	1.38	1.26	1.17	1.17	1.08	1.06
1.92	1.9	1.76	1.66	1.31	1.38	1.16	-	-	-

(b) Using the data in 3 (a), test the significant difference of the five sample means using Kruskal Wallis test.
- (a) The data in the accompanying table relate grams plant dry weight Y to percent soil organic matter X1, and kilograms of supplemental soil nitrogen added per 1,000 square meters X2:

X1	7	1	11	11	7	11	3
X2	2.6	2.9	5.6	3.1	5.2	5.5	7.1
Y	78.5	74.3	104.3	87.6	95.9	109.2	102.7

Fit a linear model of the form $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$ for the data given that

$$(X'X)^{-1} = \begin{pmatrix} 1.799597 & -0.068547 & -0.253165 \\ -0.068547 & 0.010077 & -0.001066 \\ -0.253165 & -0.001066 & 0.057079 \end{pmatrix}$$

(b) Test the significant of the percent soil organic matter in the model given that $\alpha = 0.05$.

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5. An experiment is conducted to evaluate the warping of copper plates. The response variable is a measurement of the amount of warping, and the factors of interest are temperature and the copper content of the plates. The table below lists the data from the experiment and analysis is carried out using $\alpha = 0.05$.

Temperature(-C)	Copper Content							
	40		60		80		100	
50	17	20	16	21	24	22	28	27
75	12	9	18	13	17	12	27	31
100	16	12	18	21	25	23	30	23
125	21	17	23	21	23	22	29	31

- (a) Test for the significant difference in amount of warping for the four temperatures;
 (b) Test for the significant difference in amount of warping for the four copper content of the plates.
 (c) If the null hypothesis is rejected in (a), determine the pair of temperatures that is responsible for the rejection.
6. Big sagebrush is often planted in range restoration projects. An experiment is performed to determine the effects of storage length (days) and relative humidity (percent) on the viability of seeds. Sixty-three batches of 300 seeds each are randomly divided into 21 groups of three. These 21 groups each receive a different treatment, namely the combinations of storage length (0, 60, 120, 180, 240, 300, or 360 days) and storage relative humidity (0, 32, or 45%). After the storage time, the seeds are planted, and the response is the percentage of seeds that sprout.

Humidity (Percent)	Storage Length (Days)						
	0	60	120	180	240	300	360
0%	82.1	78.6	79.8	82.3	81.7	85.0	82.7
	79.0	80.8	79.1	75.5	80.1	87.9	84.6
	81.9	80.5	78.2	79.1	81.1	82.1	81.7
32%	83.1	78.1	80.4	77.8	83.8	82.0	81.0
	80.5	83.6	81.8	80.4	83.7	77.6	78.9
	82.4	78.3	83.8	78.8	81.5	80.3	83.1
45%	83.1	66.5	52.9	52.9	52.2	38.6	25.2
	78.9	61.4	58.9	54.3	51.9	37.9	25.8
	81.0	61.2	59.3	48.7	48.8	40.6	21.0

Using $\alpha = 0.05$.

- (a) Test the null hypothesis that there is no significant difference among the three levels of relative humidity.
 (b) Test the null hypothesis that there is no significant difference among the seven levels of storage length.
 (c) Test the hypothesis about the significant of interaction between relative humidity and storage length.

Best of Luck.

Examiners: Dr. A. M. Okedoye, S. O. Ogundele

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Paper 5 of 5 — MTH 215 2022/2023

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2022/2023

Course Code/Title: MTH 215- Vector Analysis
Instructions: Answer Question 1 and any other 2 Questions. Time Allowed: 2 Hours.

QUESTION 1 (30 Marks)

(a) Define the following:

- I. Vector equation of a plane (2.5 Marks)
- II. An osculating plane & A rectifying plane (2.5 Marks)
- III. Vector valued function (2.5 Marks)
- IV. Vector Algebra and State five laws of vector algebra (2.5 Marks)

(b) Determine the equation of the plane $X(s,t) = P + sA + tB$ if $P = (1,2,-3)$, $A = (3,2,1)$ and $B = (1,0,4)$. Determine which of these points is on the plane $(6,4,6)$ and $(1,2,0)$ (5 Marks)

(c) State and prove Greens' Theorem(Gauss' Theorem) (7.5 Marks)

(d) State and prove First Fundamental Theorem of Calculus (7.5 Marks)

QUESTION 2 (20 Marks)

(a) A Particle moves along a curve $x = 2t^2$, $y = t^2 - 4t$, and $z = 3t - 5$, Find the component of velocity and acceleration at $t = 1$, in the direction $i - 3j + 2k$. (6 Marks)

(b) Given that $a = -i + 2j - 2k$, $b = 3i + 4j - 12k$, Find the angle between the vectors a and b . (4 Marks)

(c) If $A = -i + 2k$, $B = 2i + j - k$ and $C = i + 2j + 2k$. Compute each of the following:
 (i) $AX(BXC)$ (ii) $(A+B)X(A-C)$ (iii) $(AXB)X(AXC)$ (6 Marks)

(d) Find the shortest distance from the point $(1, 1, -2)$ to the surface $z = x^2 + y^2$ (4 Marks)

QUESTION 3 (20 Marks)

(a) Find the sum of the following:

- I. $PQ \cdot TQ$, PS , ST (2 Mark)
- II. VX , XY , $-ZY$, ZX (2 Mark)

(b) If $a = 3\hat{e}_p + 4\hat{e}_q + 8\hat{e}_r$, $b = 3\hat{e}_p + 2\hat{e}_q - 7\hat{e}_r$, $c = 2\hat{e}_p - 5\hat{e}_q + 2\hat{e}_r$,

- I. Find $2a - 5b + 3c$ (3 Marks)
- II. Find the components of $a - 3b + 2c$ (3 Marks)

(c) Find the area of the triangle whose two adjacent sides are given as $a = 3i - 2j + k$ and $b = 2i + 2j - 4k$. (5 Marks)

(d) Find a unit tangent vector to any point on the curve $x = t^2 - 1$, $y = 4t - 3$, $z = 2t^2 - 6t$. The position vector at any point t is $r = xi + yj + zk$. (5 Marks)

QUESTION 4 (20 Marks)

(a) Find an equation for the tangent plane to the surface $2xz^2 - 3xy - 4x = 7$ at point $(1,-1,2)$ (5 Marks)

(b) If $A = x^2yi - 2xzj + 2yzk$, Find the curl of curl of A (5 Marks)

(c) Prove that the divergence of the curl of a vector $A = 0$ (i.e. $\nabla \cdot \nabla \times A = 0$) (4 Marks)

(d) Find $\nabla \phi$ when $\phi(r)$ is a function of r alone, where $r = \sqrt{x^2 + y^2 + z^2}$ and r is a position vector. (6 Marks)